

AIX

Autonomous Driving (1)

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May, 19th, 2026

From Modular Autonomy to Tesla End-to-End

- Before Tesla V12, the dominant AV recipe was a modular stack: perception, prediction, planning, and control.
- Tesla's recent shift is historically important because it tries to replace more of that stack with learned end-to-end behavior.

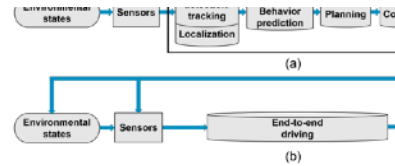
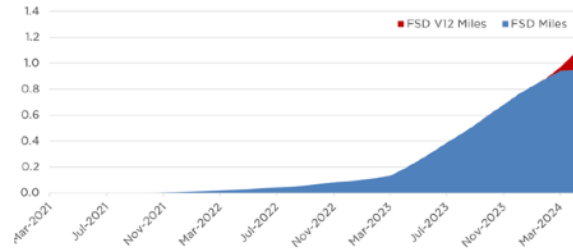


FIGURE 2: Information flow diagrams of: (a) a generic modular system, an

Core functions of a modular ADS can be summarized as:

TABLE 2: Co



Why Autonomous Driving Became an AI Problem

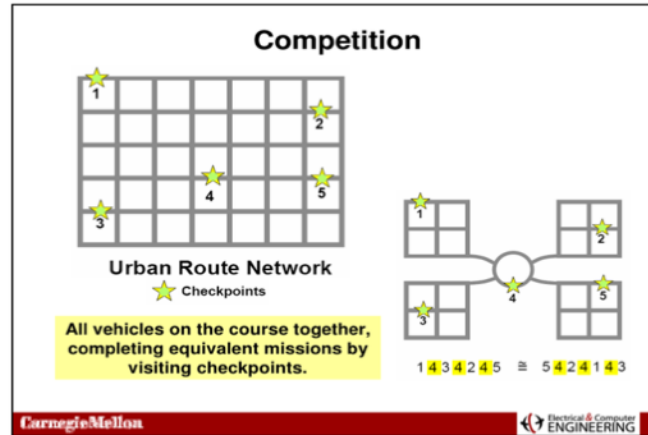
- Driving is not only a control problem; it is an open-world perception and uncertainty problem.
- Urban driving especially requires context understanding, interaction modeling, and rare-event handling.



Fig. 1: Survey at A Glance. (a) Pipeline and Methods. We define end-to-end autonomous driving as a learning-based algorithm framework with raw sensor input and planning/control output. We deepdive into 270+ papers and categorize

Before Products, There Was the Challenge Era

- Early autonomous driving matured through public robotics milestones rather than consumer deployment.
- The DARPA challenge sequence turned autonomy into a systems integration problem.



Urban Challenge Changed the Benchmark

- DARPA Urban Challenge moved the problem from off-road navigation to city driving under traffic rules.
- Intersections, blocked roads, and interactions with moving agents became central.

2007 DARPA Urban Challenge

Autonomous Driving Within City Limits

- Obey traffic laws
- Safe entry into traffic flow
- Safe passage through busy intersections
- Safe following or passage of moving vehicles
- Safe passage of a stopped vehicle
- Drive an alternate route when the primary route is blocked
- Safe U-turn (3-point turn)

Urban Challenge Already Looked Like a Stack Problem

- Even at this stage, the hard parts were mission planning, situational awareness, sensor fusion, and control.
- In other words, the later modular AV stack already had its conceptual roots here.

Summary of Challenges	
Technical Challenges	System Challenges
• Mission planning – Sequence of missions • Situational awareness and vehicle behaviors – Excessive aggression vs. caution • Perception and multi-sensor fusion • Real-time actuator control	• Develop and integrate multiple technologies • It's real – Things will go wrong • Have multiple groups and cultures work together productively • Do it all in 15 months

The Modular ADS Became the Default Blueprint

- After DARPA, most industrial ADSs converged on a modular architecture.
- The stack was usually decomposed into localization, perception, prediction, planning, and control.

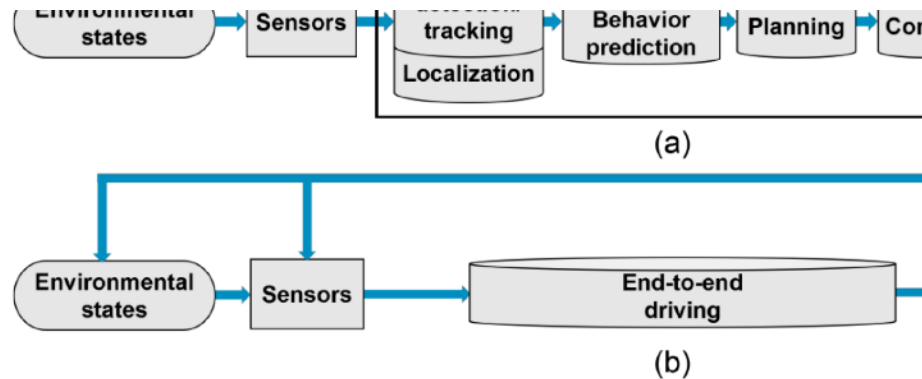


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What a Production Stack Actually Contains

- Industrial autonomy is not one network; it is a coordinated set of subsystems.
- Cruise's 2018 safety report makes this modular logic very explicit.

Autonomous driving is a complex system that requires a combination of systems that work safely together, including:



Cruise-Style Autonomy Is Modular by Construction

- The Cruise AV was designed around sensing, interpretation, and decision modules, not a single end-to-end policy.
- The sensor suite and the stack were engineered together to support structured autonomy.

HOW THE CRUISE AV OPERATES

Let's look at three of these elements — **Perception**, **Planning** and **Controls** — to showcase how the Cruise AV senses its environment and makes driving decisions.

Allows safe operation based upon both what the sensors "see," as well as what may be hidden from view.

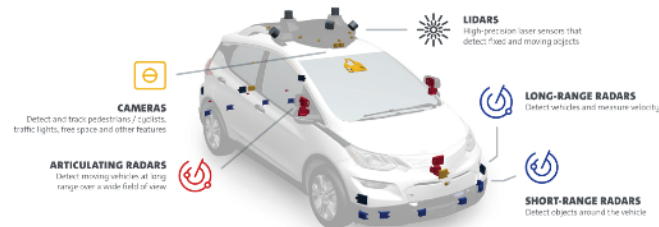
In our self-driving vehicle, **Perception** "sees" by using sensors to monitor its environment. The sensors feed information to the computer that combines the sensor data with high-definition map data to localize the vehicle. Perception detects and classifies

objects, determines their location and provides their speed and direction. It builds a three-dimensional model of the world that keeps track of important objects. Perception also predicts the objects' future motion — pedestrians and trucks have different predicted movements. Using the three-dimensional model and map data, Perception determines free, drivable space around the vehicle.

Perception identifies other environmental uncertainties. For example, with knowledge of its location, Perception knows where it must look for moving objects. If its view is blocked, Perception will flag that area as unknown. If an object is hard to see because of rain or fog, or because it is hidden behind a truck, the computer brain knows that and adjusts its decision-making and performance accordingly.

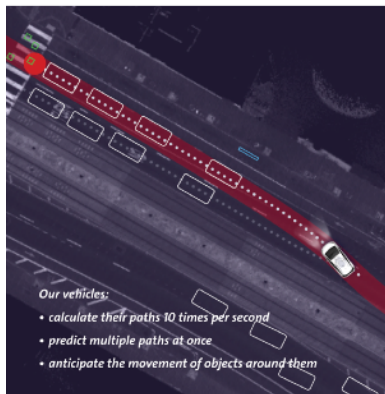
This allows prudent decision-making and operation based upon both what the sensors "see," as well as what may be hidden from view.

To perform Perception functions, the vehicle has five LiDARs, 16 cameras and 21 radars. Their combined data provides sensor diversity, allowing Perception to see complex environments. Our LiDARs, radars



Perception, Planning, and Control Were Separated on Purpose

- Perception estimated the scene, planning chose a safe path, and control executed it.
- This separation improved engineering clarity even if it limited global learning.



and cameras all scan both long and short range with views 360 degrees around the vehicle. We start with LIDAR, which provides highly precise feedback using laser measurements for both fixed and moving objects. Radar is complementary to LIDAR because it uses electromagnetic pulse measurements and can see solid objects that have low light reflectivity. We use both LIDAR and radar inputs for measuring the speed of moving objects, allowing quick, confident determinations of speed. Cameras are also complementary to LIDAR because they measure the light intensity reflected off an object

from objects, providing rich detail of the object. We combine LIDAR and camera data for classifying and tracking objects, making high-confidence determinations more quickly. This helps, for example, identify pedestrians, vehicle types and road details such as lane lines, construction zones and signage. Our complementary set of long-range sensors track high-speed objects, such as oncoming vehicles, and the short-range sensors provide detail about moving objects near the vehicle such as pedestrians and bicycles.

With an understanding of space and time, the car plans its path

Within the computers' operations, **Planning** determines the desired vehicle behavior. It accounts for road rules and plans routes for the car to travel from trip origin to destination. It chooses routes to optimize efficiency and safety and to route the car only on streets within its capabilities.

Planning operations are based upon vehicle location, other road users' predicted actions, traffic controls, road markings, rules of the road and other external factors. Planning identifies multiple paths per second, and constantly chooses the best one to meet changing road conditions and events.

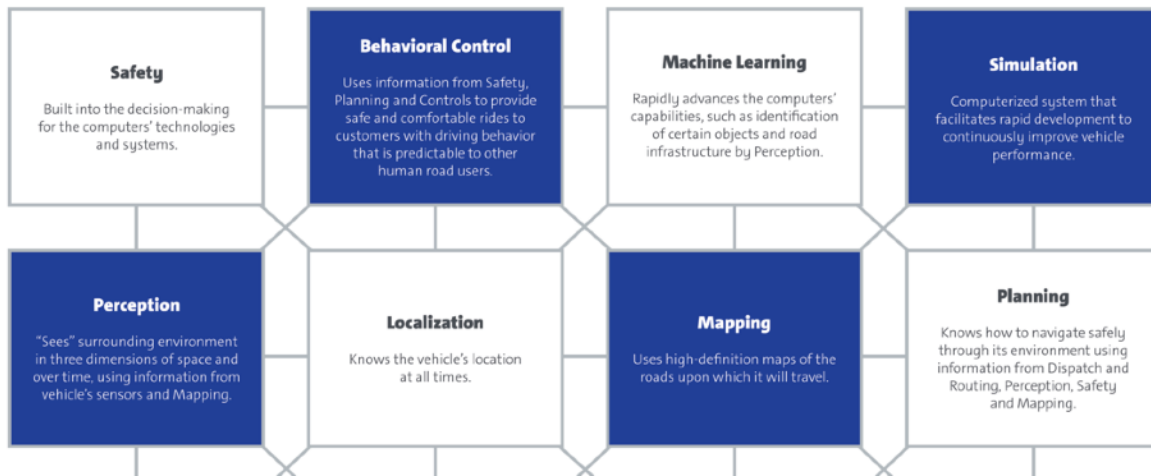
If something unexpected happens, Planning has multiple backup plans. For example, while preparing to change lanes to turn right at an intersection, another vehicle may aggressively cut into the destination lane, making the lane change unsafe. Planning would already have an alternative route planned; for example, the vehicle could go around the block instead of blocking its current lane while waiting for an opening to change lanes.

The **Controls** function implements the final path from Planning, converting its commands for the actuators that control the steering, throttle, brake and drive unit. We've designed the Controls function to give the self-driving system full vehicle maneuverability complete with stability, traction and anti-lock brake system fully active.

Where Machine Learning Entered First: Perception

- In early industrial AV systems, machine learning mostly entered through perception.
- Detection, classification, segmentation, and tracking were the first big AI wins in driving.

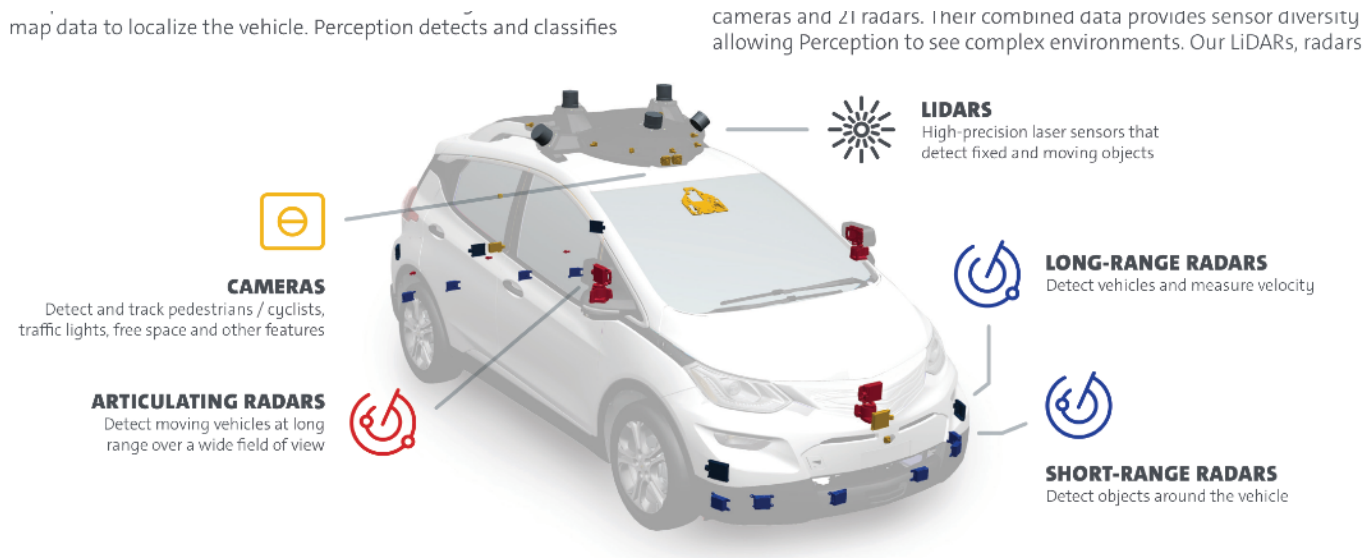
Vehicle and make the driving decisions that safely transport passengers. No one technology makes this happen. Instead, the computers use a combination of systems that work safely together, including:



Cruise system diagram — Perception and ML blocks

Sensor Fusion Was the Main Design Question

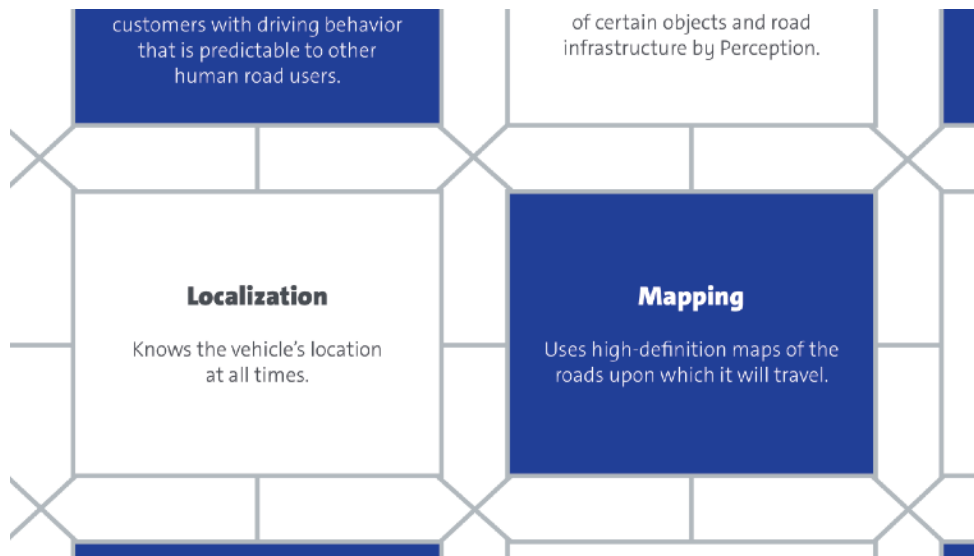
- The key system question was often how to combine cameras, LiDAR, radar, and maps.
- This is very different from asking what one learned representation should contain.



Cruise sensor-annotated vehicle

HD Maps and Localization Were First-Class Modules

- The vehicle had to know where it was before it could reason about what to do.
- Localization and mapping were therefore explicit modules, not implicit latent variables.



Mobileye and RSS Represent the Rule-Based Safety View

- Mobileye's RSS formalizes safe driving as transparent rules and proper responses.
- This view treats safety as a provable envelope around a driving policy.



Responsibility-Sensitive Safety (RSS):

Forging a Universal Definition for Autonomous Vehicle Safety

Autonomous vehicles (AVs) have life-saving potential. But without a verifiable way to demonstrate their ability to drive safely, these vehicles will never get a license to drive.

Driving safely – and a universal definition of what that means – is the final and most crucial hurdle standing in the way of our autonomous future. Technology leaders, automakers and government bodies must collaboratively establish a methodology and standard for the assessment and verification of an AV to drive safely and thus be worthy of a license to drive.

To this end, Intel has contributed an open, technology-neutral framework as a starting point for the industry to align on what it means for an AV to drive safely: Responsibility-Sensitive Safety (RSS).

What is Responsibility-Sensitive Safety (RSS)?

RSS in One Sentence: Safe Distance and Proper Response

- RSS defines safety as a set of provable constraints rather than a learned feeling of what is safe.
- The key pedagogical point is that safety is treated as an explicit envelope around any proposed behavior.



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What is Responsibility-Sensitive Safety (RSS)?

Why Industry Wanted a Safety Envelope

- A learned policy may be useful, but regulators and engineers still need a way to say when an action is definitely unsafe.
- This is why rule-based safety layers remained attractive even as ML entered perception.

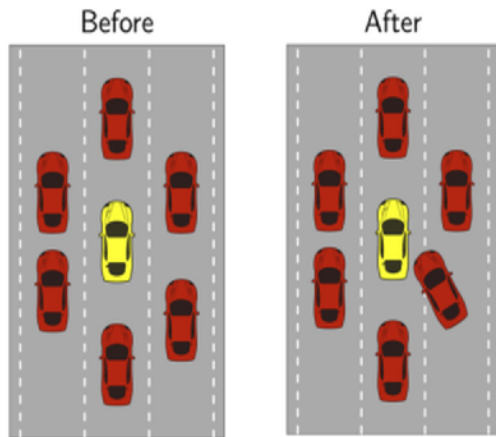
performed offline⁵ through the use of large ground truth data and abundant sensing modalities that ensure there are at least two independent observers to simplify validation for each subsystem. A detailed discussion is in Section 5.2. Planning on the other hand, presents unique challenges. The planning process and decisions should depend on the actions and responses of other agents. During testing, we do not do so on a closed track but rather in the real world. The perception making capabilities can be sufficiently evaluated. In the next section, we discuss the safety of planning, and in Section 3 we describe our RSS architecture.

Approaches to Claims on Safe AV Decision Making

Approaches to claiming the safety of AV are currently being promoted in the industry. These include simulation based testing and proprietary approaches. The simulation approach is based on a statistical argument attempting to show that a system is safe for a large number of drivers. This approach is problematic because of the sheer volume of data required to show enough statistical evidence that a claimed probability of error

RSS Separates Safety Rules from Learned Decision Makers

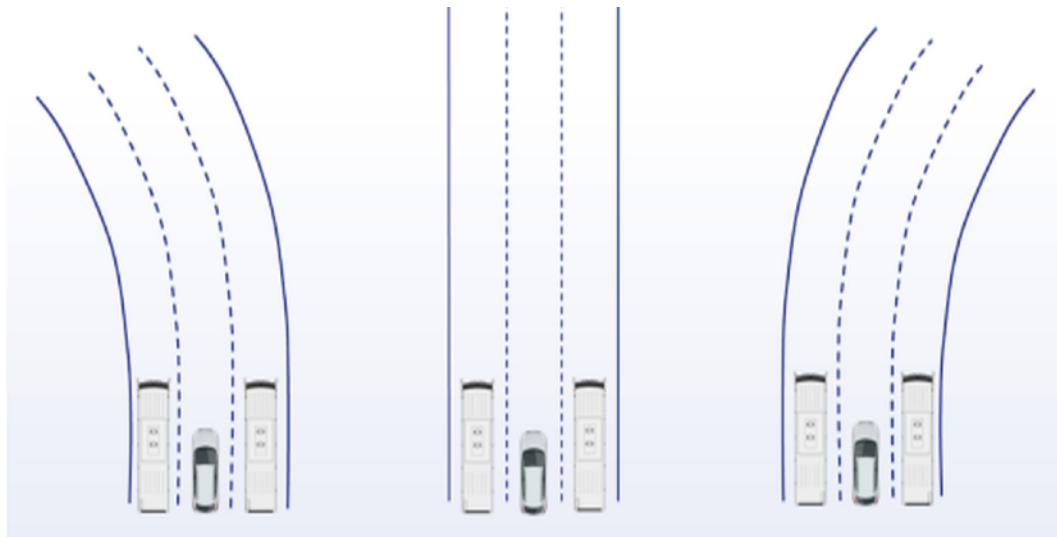
- In the RSS view, learned models may help, but safety cannot depend only on opaque behavior.
- The central idea is that a system must know when an action is unsafe regardless of how it was proposed.



Mobileye safety architecture paper, Figure 1

Rules and Learning Coexisted Before End-to-End

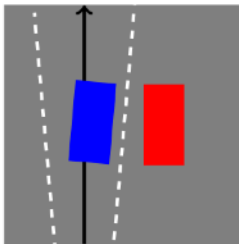
- Mobileye's later architecture explicitly combines learned subsystems with a guarded fusion logic.
- This is not 'rules or learning'; it is 'learning inside a safety-constrained stack.'



Mobileye SDS: lane-geometry fusion figure

Hybrid Systems: Not Rules or Learning, but Rules plus Learning

- Before large-scale end-to-end training, most industrial systems used learning where it clearly helped and rules where accountability mattered most.
- This hybrid strategy explains why modular AV systems survived for so long.

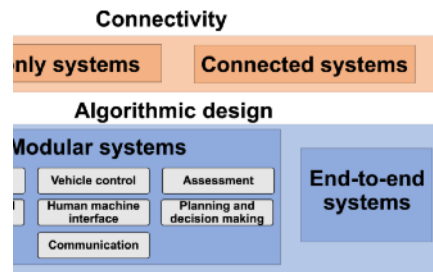


: Changing lane width. Although the red car drives in parallel to the lane's center (black arrow), lateral movement towards the lane. The blue car, although getting further away from the lane's center position w.r.t. the lane boundary.

Why Modular Systems Dominated for So Long

- Modular systems are easier to debug, validate, and assign to separate engineering teams.
- Their interpretability was a major industrial advantage before large-scale end-to-end learning became plausible.

level system architectures



A high level classification of automated driving architectures

Reliability certification is another task yet to be completed. ADAS need to be designed with high

Weakness 1: Error Propagation Across Modules

- Once the stack is split into modules, downstream stages inherit upstream mistakes.
- Locally strong modules can still produce globally weak driving behavior.

Yu Qiao¹, Hongyang Li^{1†}

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³ SenseTime Research

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/OpenDriveLab/UniAD

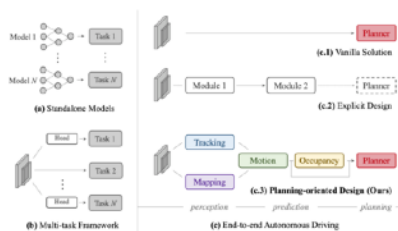


Figure 1. Comparison on the various designs of autonomous driving framework. (a) Most industrial solutions deploy separate models for different tasks. (b) The multi-task learning scheme shares a backbone with divided task heads. (c) The end-to-end paradigm unites modules in perception and prediction. Previous attempts either adopt a direct optimization on planning in (c.1) or devise the system with partial components in (c.2). Instead, we argue in (c.3) that a desirable system should be planning-oriented

What Error Propagation Looks Like in Practice

- If perception makes a small mistake, prediction inherits it, planning optimizes on top of it, and control executes a flawed plan.
- This means locally good modules can still create globally bad driving.

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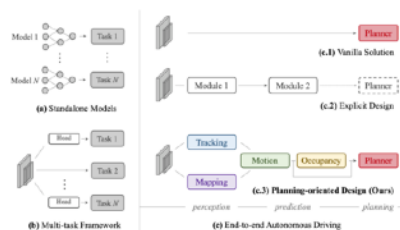
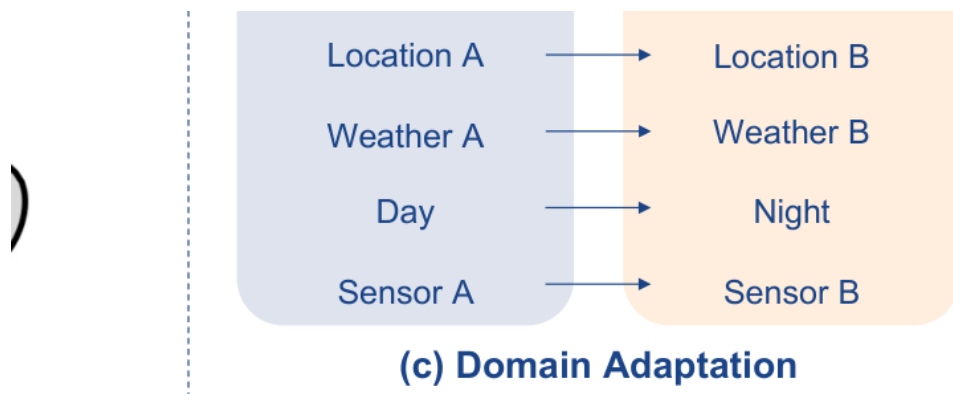


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Weakness 2: Long-Tail and Corner Cases

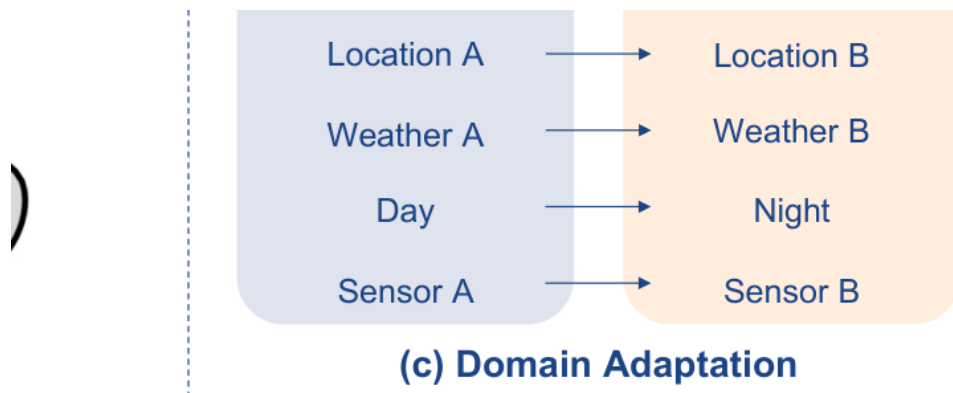
- Rare events expose the limits of hand-crafted rules faster than average scenarios do.
- Adding more heuristics often increases system complexity faster than coverage.



arise in relation to dataset distribution discrepancies, scenarios, and domain shift in locations, weather, *etc.*

Corner Cases Are a Coverage Problem, Not Just a Performance Problem

- Rare events matter because safety is determined by coverage of unusual cases, not only by average-case accuracy.
- Each extra rule may cover one new case while making the whole stack harder to maintain.

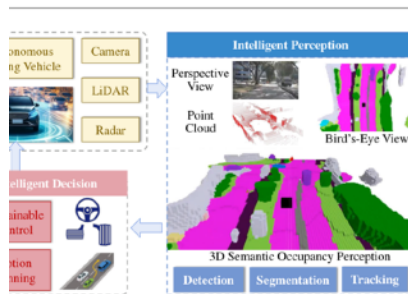


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Weakness 3: Stitched Outputs Do Not Form a World Model

- Object lists, lane lists, and freespace masks do not automatically compose into a coherent world state.
- Occlusion, uncertainty, and interaction are then pushed downstream as engineering burdens.

ion, Multi-Modal Data.



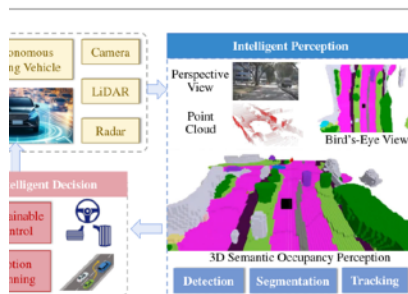
: **Autonomous driving vehicle system.** The sensing data from camera, LiDAR, and radar enable the vehicle to intelligently perceive its surroundings. Subsequently, the intelligent decision module generates control and plan-driving behavior. Occupancy perception surpasses other perception based on perspective view, bird's-eye view, or point clouds, in terms of understanding and density.

In academia and industry, occupancy perception for holistic scene understanding poses a meaningful impact. On

Why an Object List Is Not Yet a World Model

- A list of cars, lanes, and free-space masks does not automatically tell the planner what the hidden geometry of the scene is.
- Occlusion and uncertainty are exactly where modular outputs stop being enough.

ion, Multi-Modal Data.



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Learning Starts Moving Beyond Boxes

- Once the field asks for richer scene structure, perception must model more than discrete objects.
- This is where occupancy-style representations become historically important.

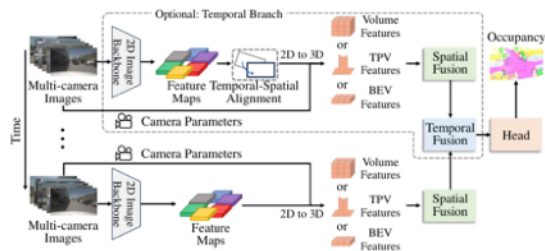


Figure 5: **Architecture for vision-centric occupancy perception:** Methods without temporal fusion [31, 32, 36, 38, 76, 81, 82, 83, 87, 116]; Methods with temporal fusion [4, 8, 10, 56, 80, 85].

ages the results of the 2D branch to enhance the density of the output from the 3D branch. Ablation studies report that this straightforward and direct information fusion strategy can yield

Occupancy Intuition: Predict the Space, Not Only the Classes

- Occupancy asks which 3D cells are filled, empty, or semantically typed, rather than only asking which known objects exist.
- This makes unknown obstacles and irregular geometry easier to represent.

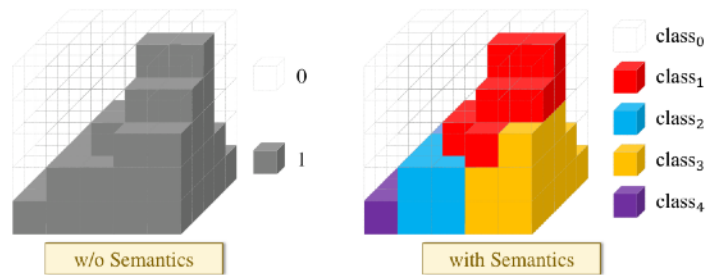


Figure 3: **Illustration of voxel-wise representations with and without semantics.** The left voxel volume depicts the overall occupancy distribution. The right voxel volume incorporates semantic enrichment, where each voxel is associated with a class estimation.

Occupancy Formalizes Space Instead of Object Lists

- Occupancy discretizes 3D space and predicts which cells are filled, empty, or semantically labeled.
- This gives the model a geometry-first world view.

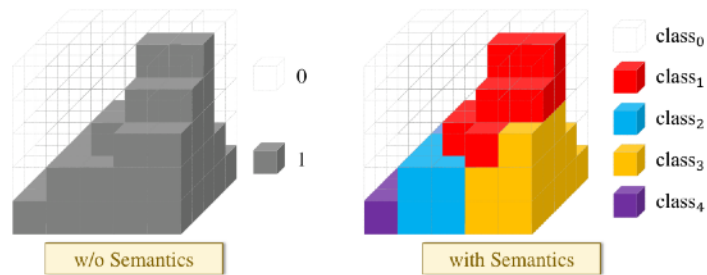
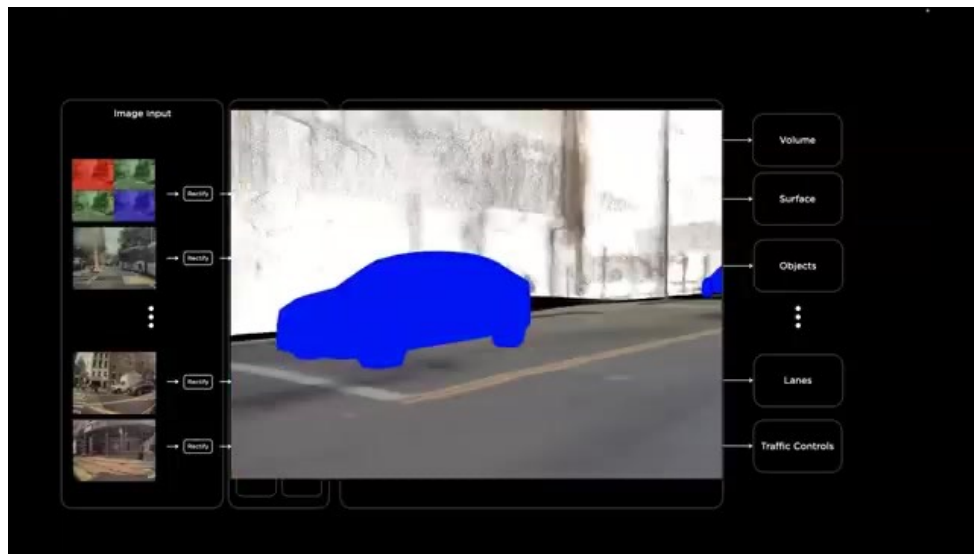


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From Occupancy to Data Engine

- A richer representation is not enough by itself; it must also be supported by scalable labeling, mining, and retraining.
- This is the bridge from representation innovation to Tesla's later data-engine story.



Planning-Oriented Learning Is a Transitional Step

- The next idea was not yet full end-to-end driving, but making representations more planning-aware.
- This aligns the learned representation with the final driving objective more directly.

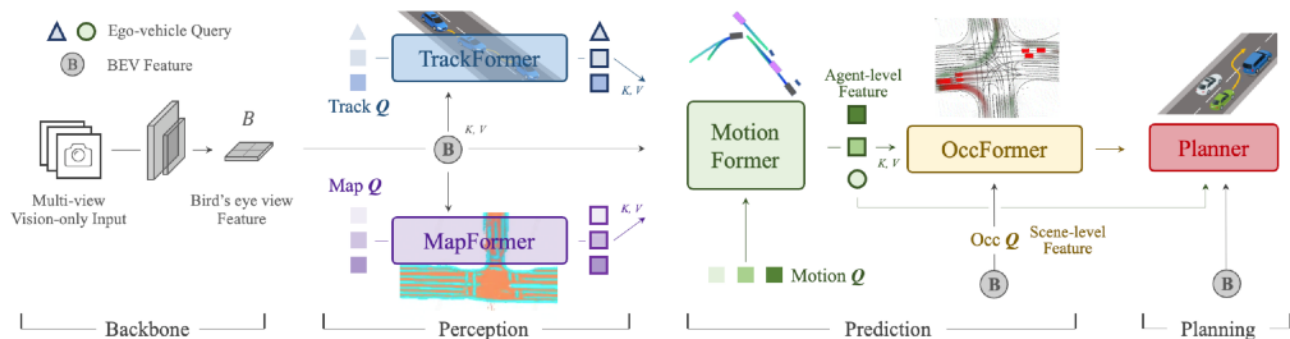


Figure 2. **Pipeline** of Unified Autonomous Driving (UniAD). It is exquisitely devised following planning-oriented philosophy. Instead of a simple stack of tasks, we investigate the effect of each module in perception and prediction, leveraging the benefits of joint optimization from preceding nodes to final planning in the driving scene. All perception and prediction modules are designed in a transformer decoder structure, with task queries as interfaces connecting each node. A simple attention-based planner is in the end to predict future waypoints of the ego-vehicle considering the knowledge extracted from preceding nodes. The map over occupancy is for visual purpose only.

The Field Shifts from Pipeline Design to Representation Learning

- The historical center of gravity moves from handcrafted interfaces to learned world representations.
- This is the intellectual background for Tesla's later end-to-end story.

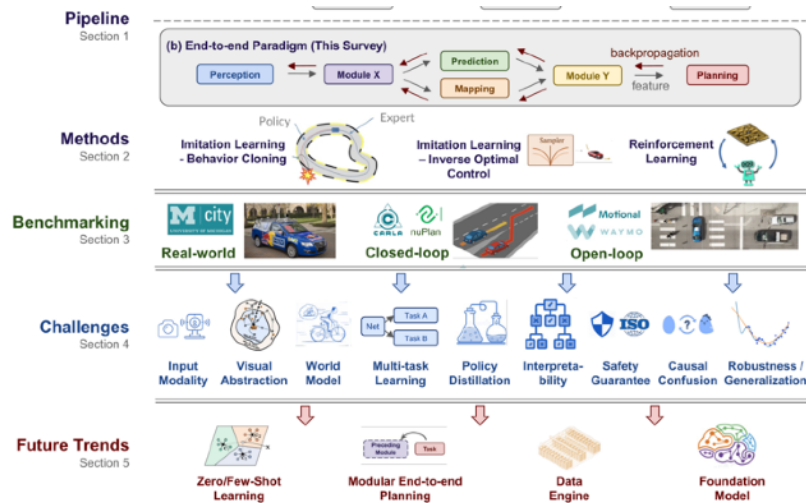


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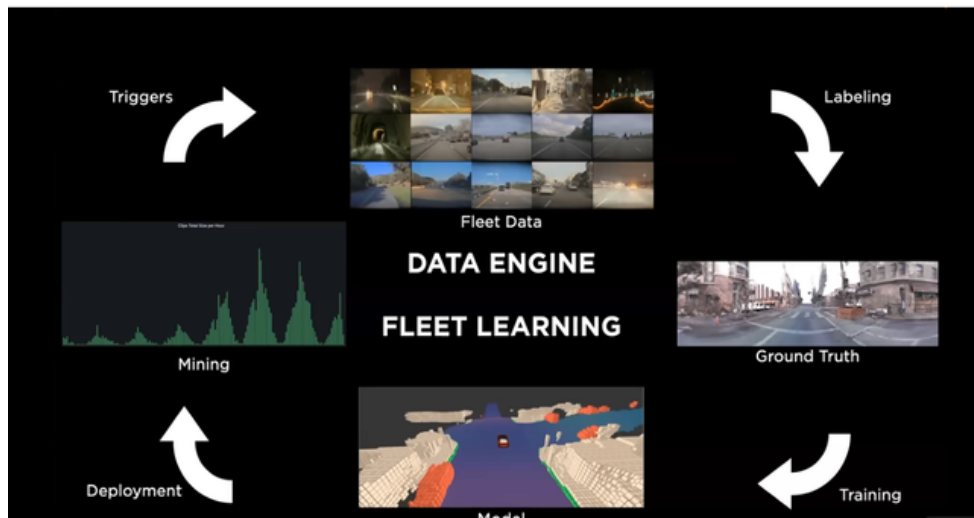
Why Camera-First Learning Became Attractive

- If fleet video becomes the main data asset, cameras gain strategic value beyond sensor cost.
- A camera-first system is easier to scale across fleets than a sensor-rich stack built for one vehicle class.



Fleet Data Changes What Can Be Learned

- Once millions of cars collect video, the bottleneck shifts from rules to data selection, labeling, and training.
- This is where Tesla's historical path diverges from the earlier industry playbook.



Software 2.0 Reframes the Problem as Learning

- The competitive advantage moves from writing more rules to running a better data engine.
- This is the conceptual bridge from modular autonomy to Tesla's end-to-end agenda.

CORE TECHNOLOGY

Artificial Intelligence Software and Hardware

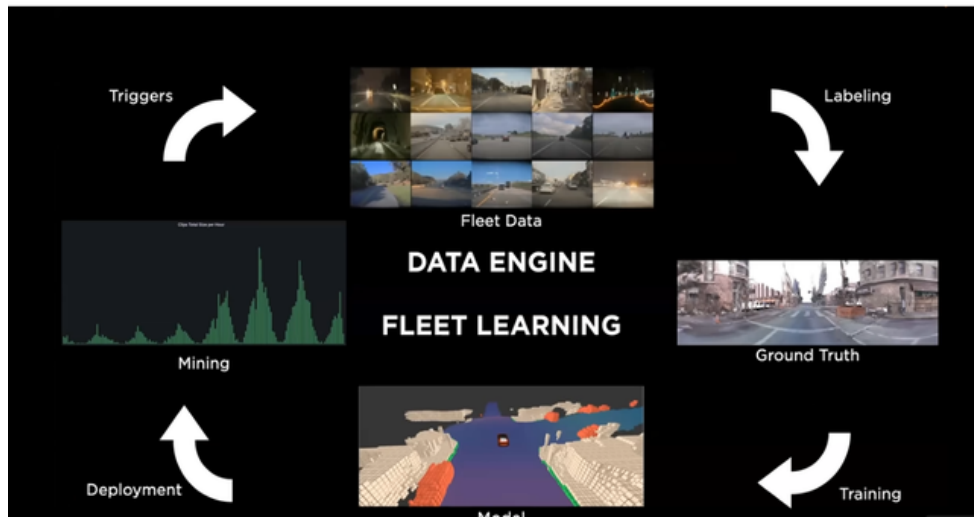
We have been investing in the hardware and software ecosystems necessary to achieve vehicle autonomy and a ride-hailing service. We believe a scalable and profitable autonomy business can be realized through a vision-only architecture with end-to-end neural networks, trained on billions of miles of real-world data. Since the launch of FSD (Supervised) V12 earlier this year, it has become clear that this architecture long pursued by Tesla is the right solution for scalable autonomy. To further improve our end-to-end training capability, we will continue to increase our core AI infrastructure capacity in the coming months. In Q1, we completed the transition to Hardware 4.0, our latest in-vehicle computer with increased inference processing power and improved cameras.

Vehicle and Other Software

We rolled out FSD (Supervised) with a 30-day free trial to eligible cars in U.S. and Canada. FSD (Supervised) can change lanes, select forks to follow routes, navigate

Data Engines Matter as Much as Model Architecture

- In large-scale autonomy, the closed loop of mining, labeling, training, and redeployment is part of the method.
- The 'model' is therefore a system, not just a network.



From Modular Autonomy to Tesla End-to-End

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- Tesla's recent shift is historically important because it tries to replace more of that stack with learned end-to-end behavior.

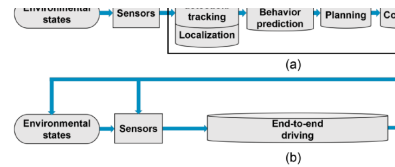
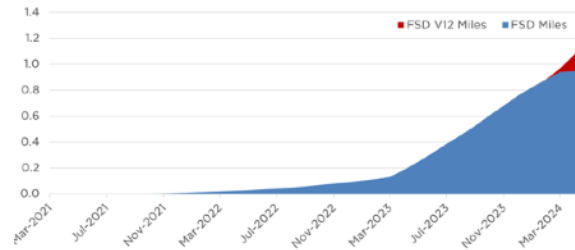


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Why Tesla's Shift Needed Three Things at Once

- End-to-end driving becomes plausible only when representation, fleet-scale data, and training compute are all strong enough together.
- This is why the 2024 Tesla story should be taught as a system-level change, not only as a model change.

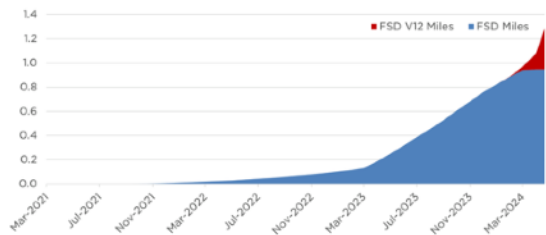
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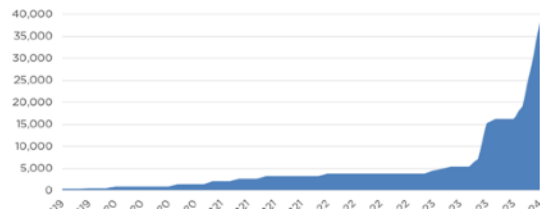
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Battery, Powertrain & Manufacturing



Cumulative miles driven with FSD Beta (billions; as of Apr-21-24)



Historically, What Changed the Most?

- The core question changed from 'how do we engineer the right stack?' to 'what world representation can the system learn?'
- That is the broad historical transition students should remember before studying Tesla in detail.



Fig. 1: Survey at A Glance. (a) Pipeline and Methods. We define end-to-end autonomous driving as a learning-based algorithm framework with raw sensor input and planning/control output. We deepdive into 270+ papers and categorize

Bridge to Deck 2: From Occupancy to Foundation Models

- Tesla's 2022 occupancy pivot is best seen as the bridge between classic modular AVs and later foundation-model narratives.
- The next deck should therefore start with occupancy and then move into world-model / foundation-model reasoning



Foundation Models for Autonomous Driving Systems: An Initial Roadmap

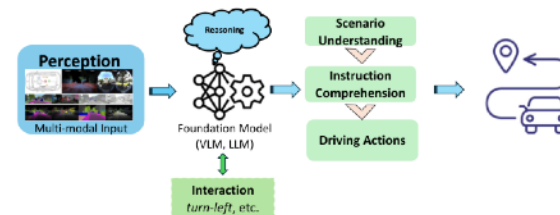


Fig. 2. A typical pipeline of enhancing autonomous driving leveraging FMs.

have explored how foundation models can advance autonomous driving more broadly [64, 216, 227]. While these works provide valuable insights and highlight important challenges and opportunities, most focus on specific topics or model